

UNIT I – INTRODUCTION AND FUNDAMENTAL DATA STRUCTURES

Linked Lists – single, double, circular,

Stack Implementation (Array & Linked List)

Applications: Expression evaluation, syntax checking (Balanced Parenthesis)

Queue Implementation (Array & Linked List)

Circular queue

Applications in scheduling, Undo-redo systems, and task queue management
(Refer Lab Program)

UNIT II – TREES AND THEIR APPLICATIONS

Expression Tree

Binary Search Trees

AVL Trees

Tree Traversals

Priority Queues (Binary Heap).

Trie (prefix tree)

Segment Tree / Binary Indexed Tree (introductory level),

Huffman coding.

UNIT III – GRAPHS AND NETWORKED SYSTEMS

Traversals – BFS, DFS

Shortest-Path Algorithms- Dijkstra's Algorithm,

Minimum Spanning Tree: Prim's Algorithm, Kruskal's Algorithm

Dependency resolution (course planning, software builds)

UNIT IV – SEARCHING, SORTING, AND PROBLEM SOLVING TECHNIQUES

Searching: Binary search and variants

Sorting algorithms: Quick sort, Merge sort using Divide and conquer technique

Heap sort

Greedy algorithms –Coin change, activity selection

Backtracking – N-Queens

Dynamic Programming: Knapsack, Longest Common Subsequence (LCS),
Matrix Chain Multiplication

UNIT V – HASHING, MEMORY MANAGEMENT, AND REAL-WORLD APPLICATIONS

Collision handling

Disjoint Set (Union-Find) – for connected components,

Memory management techniques and garbage collection logic

Real-world applications:: Real-time search (e.g., Google autocomplete) Also in U.II

Social network graphs (Facebook friend suggestions) Also in U.III
Fraud detection using graph cycles
Page replacement